MIRZA ABDUL REHMAN BIN ZAFAR

Electrical Engineering Student | Electronics

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ABOUT ME

Electrical Engineering student (CGPA: 3.21/4.00) specializing in Electronics, Embedded Systems, and Automation with hands-on experience in circuit design, control systems, IoT solutions, and hardware-software integration. Proficient in MATLAB/Simulink, Proteus, Multisim, Arduino, ESP32, C/C++, and embedded system development. Skilled in microcontrollers, PLC fundamentals, testing & validation, troubleshooting, and technical documentation. Former Teaching Assistant for Applied Physics and current President of IEEE Student Branch CFD, seeking internship opportunities in electronics, embedded systems, automation, control systems, and hardware design to contribute to real-world engineering projects.

EDUCATION

FAST NATIONAL UNIVERSITY (NUCES), CHINIOT-FAISALABAD | 2023 – 2027

Bachelor of Science in Electrical Engineering

- Status: 6th Semester (Expected Completion: June 2027) | CGPA: 3.21/4.00
- Relevant Coursework: Power Electronics, Electronic Circuit Design, Feedback Control Systems, Electrical Network Analysis

TECHNICAL SKILLS

Electrical & Power Systems

- Electrical equipment testing, inspection, and validation
- Power distribution and load management fundamentals
- Electrical protection, grounding, and earthing systems
- UPS systems, backup power, and fault analysis

Embedded & Automation Systems

- Arduino, ESP32, Raspberry Pi microcontrollers
- Hardware-software integration and interfacing
- Real-time monitoring and IoT system design
- Basic PLC logic, automation, and control concepts

Programming Languages

- C, C++, Embedded C, Assembly

Engineering & Simulation Tools

- MATLAB/Simulink
- Proteus
- TIAPortal
- TriLogi
- Arduino IDE
- TinkerCad
- Multisim
- KiCad

EXPERIENCE

Teaching Assistant - Applied Physics

FAST NUCES | Aug 2024 – Jan 2025

- Assisted undergraduate students with physics laboratory experiments, measurements, and numerical problem-solving.
- Delivered tutorial and practical support to explain engineering physics concepts and laboratory procedures.
- Guided students in data interpretation, troubleshooting, and report preparation to strengthen conceptual understanding.

ACADEMIC PROJECTS

Automatic Transfer Switch with Wi-Fi Mobile Control

- Designed and implemented an ESP32-based ATS system for automatic power source switching.
- Integrated mobile application control for remote monitoring, system status tracking, and operation.
- Developed real-time switching logic to improve response, reliability, and backup power continuity.

Home Automation System

- Developed an IoT-based smart home automation platform using sensors, microcontrollers, and wireless control.
- Enabled remote device monitoring and control through hardware-software integration.

Additional Engineering Projects

- Automatic Traffic Light Control System
- Fire Alarm Detection System
- Ultrasonic Sensor-Based EV Obstacle Avoidance System
- Mobile Signal Detection System
- FM Receiver using Operational Amplifiers
- Temperature-Controlled DC Fan
- RC Car Platform Development
- Online Airplane Reservation System (Object-Oriented Programming)

LEADERSHIP & ACTIVITIES

President – IEEE Student Branch CFD (2026 Present)

- Led technical initiatives, engineering workshops, and student engagement activities.

Student Young Professionals Coordinator – IEEE SB CFD

- Coordinated industry engagement and professional development activities.

Vice Head Logistics – DAIRA '26

- Managed operational planning and event execution for university activities.

PROFESSIONAL SKILLS

- Microsoft Office suite (Word, Excel, PowerPoint) • Communication and Technical Documentation • Teamwork and Collaboration • Leadership and Event Management • Report Writing and Documentation

AREA OF INTEREST

- Electronics • Analog Electronics • Control Systems • Circuit Design • Robotics Systems